Pang-Wei Huang (黃邦維)

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Education

University of Illinois Urbana-Champaign — **Urbana, IL, USA**
Master of Computer Science (MCS) — *Aug. 2026 – Expected Jan. 2028*

National Taiwan University — **Taipei, Taiwan**
B.S. in Computer Science and Information Engineering — *Jun. 2025*

Technical Skills

Languages: C, C++, C#, Python
Technologies: Linux, Git, Docker, WPF/XAML, GitLab CI/CD, GitLab Runner, OpenMP, Pthreads

Work Experience

Alifecom Technology — **Taiwan**
Assistant Engineer, R&D Department — *Aug. 2025 – Jan. 2026; Jun. – Jul. 2026*

- Developed and maintained **3–4 WPF (C#/XAML) desktop applications** for wireless equipment control and RF testing; adapted legacy GUIs to new hardware and refactored duplicated code to separate UI and device-control logic.
- Automated RF procedures by integrating board communication via I^2C/Arduino/serial with Keysight MXA, MXG, and VNA instruments via **SCPI over TCP**.
- Enabled factory operators to run previously engineer-operated RF tests at production scale, with an *estimated* **3–5× increase in testing throughput**.
- Implemented an **SCPI command parser** in a customer-facing GUI for programmatic control of a wireless base-station simulator.
- Deployed **GitLab CE** on Ubuntu and configured a Windows **GitLab Runner** for WPF builds; implemented VM snapshots and scheduled backups to NAS with retention policies.
- Evaluated and demonstrated **AI coding agents** (OpenCode, Claude Code, Codex), contributing to team adoption amid hiring constraints and increasing customer demand.

Selected Projects

Can't Stop: Reinforcement Learning Agents | GitHub | Demo — *2026*

- Built a reproducible simulator and trained **Ridge, Double DQN, and PPO** agents to learn Stop/Continue decisions in Can't Stop, holding dice pairing fixed to isolate the decision under study.
- In seeded, seat-swapped evaluation, our agent **outperformed all open-source Can't Stop implementations evaluated**; recorded game-level results and used paired-bootstrap analysis.
- Exported lightweight policies for **local JavaScript inference**, enabling browser gameplay without a Python runtime or inference server.

Linux Kernel Programming — UIUC CS 423 — *Fall 2026*

- Built a Linux kernel module for **process CPU monitoring** using ProcFS, kernel timers, workqueues, linked lists, slab allocation, and synchronization primitives.
- Implemented a kernel-module **preemptive Rate-Monotonic Scheduler** with task registration/yield, admission control, a dispatcher thread, wake-up timers, and priority-based preemption; passed course tests for both projects.

Wireless Communications Experiment Reproduction | GitHub — *Jan. 2024*

- Reproduced published wireless communications experiments with **C++ simulations**; analyzed and visualized results in Python, validating against theoretical predictions and independent randomized trials.